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import socket
import sys

SERVER_IP = '10.0.0.1' # Mininet topology IP
PORT = 5000

def evaluate_connection():
    print("=====")
    print("  CLIENT-SERVER EVALUATION SCRIPT  ")
    print("=====")

    client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    client_socket.settimeout(3.0)

    try:
        print(f"\n[*] Connecting to Server at {SERVER_IP}:{PORT}...")
        client_socket.connect((SERVER_IP, PORT))
        print("[✓] SUCCESS: Connected to server successfully!")

        test_username = "EvalUser\n"
        client_socket.sendall(test_username.encode('utf-8'))
        print("[✓] SUCCESS: Username sent successfully.")

        client_socket.close()
        print("\n=====")
        print(" RESULT: PASSED (Socket connection is working!)")
        print("=====")
        return True

    except socket.timeout:
        print("\n[✗] ERROR: Connection timed out. Make sure server.py is running.")
    except ConnectionRefusedError:
        print("\n[✗] ERROR: Connection refused. Check if server IP and Port are correct.")
    except Exception as e:
        print(f"\n[✗] ERROR: {e}")

    print("\n=====")
    print(" RESULT: FAILED")
    print("=====")
    return False

if __name__ == "__main__":
    evaluate_connection()

```